

— Abdominal walls

Objectives

- Anterolateral abdominal wall
 - Posterior abdominal wall
 - The diaphragm
 - The lumbar plexus
 - The aorta
-
- The inferior vena cava

DR.RASHA ALSHALI

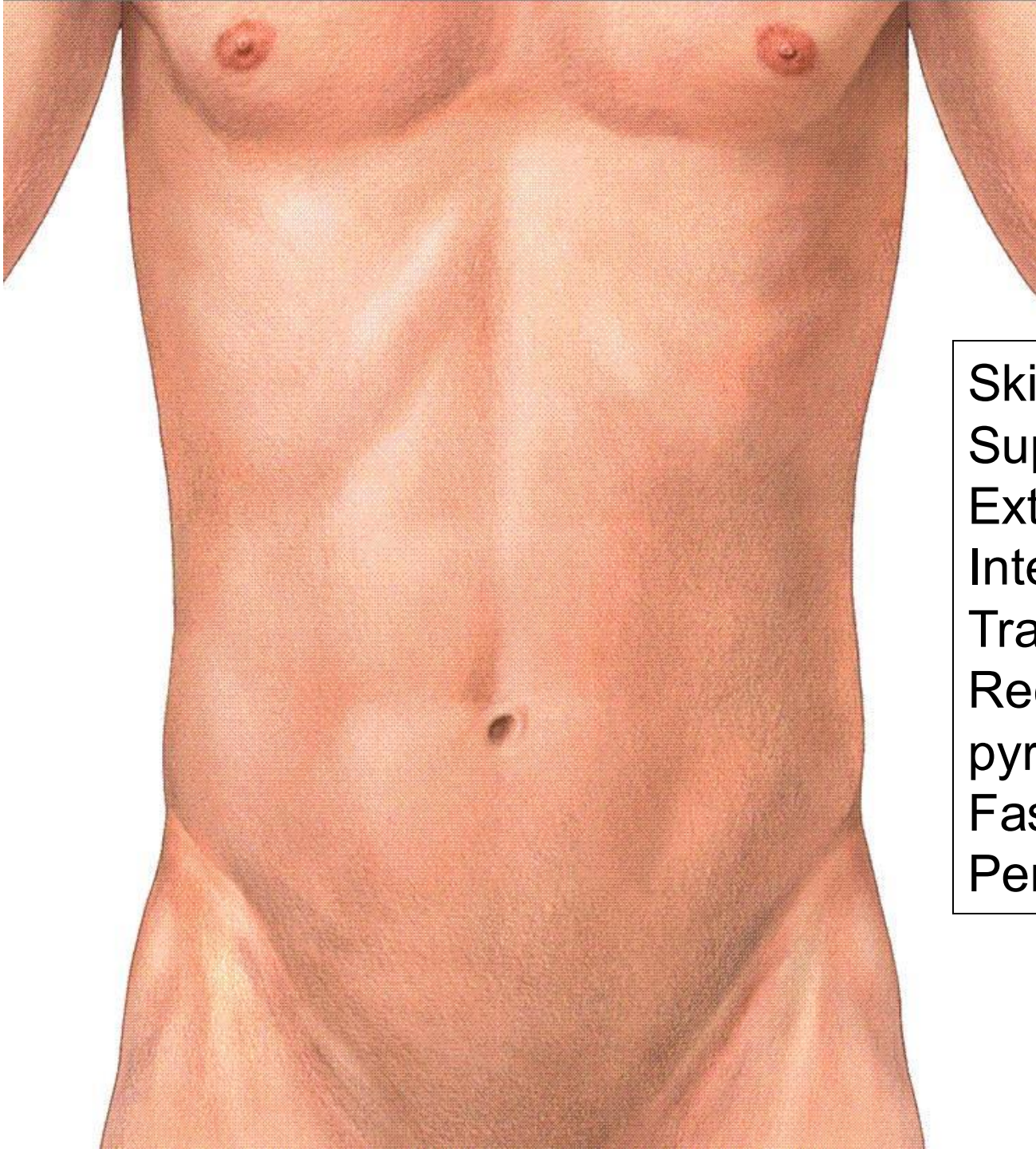
Associate professor

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KAU

The anterolateral abdominal wall

1. Recognize the different layers of anterolateral abdominal wall.
2. Recognize different muscles of anterolateral abdominal wall (origin, insertion, nerve supply & action)
3. Understand the Rectus sheath (formation & its contents).
4. Describe the anatomy of the Inguinal canal (formation & its contents).



Layers of the anterolateral abdominal wall

Skin.

Superficial fascia.

External oblique muscle.

Internal oblique muscle.

Transversus abdominis muscle.

Rectus abdominis muscle & its sheath.

pyramidalis muscle.

Fascia transversalis.

Peritoneum.

The layers of anterolateral abdominal wall

Laterally:

- 1- Skin.
- 2- Superficial fascia.
- 3- External oblique muscle.
- 4- Internal oblique muscle.
- 5- Transversus abdominis muscle.
- 6- Fascia transversalis.
- 7- Peritoneum.

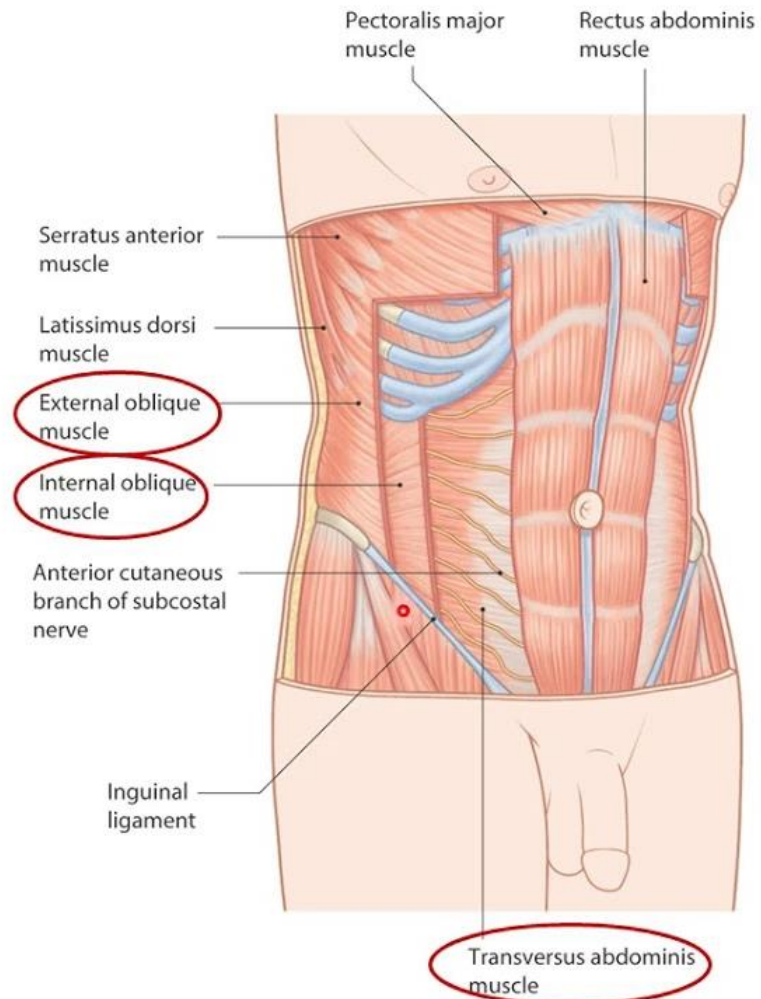
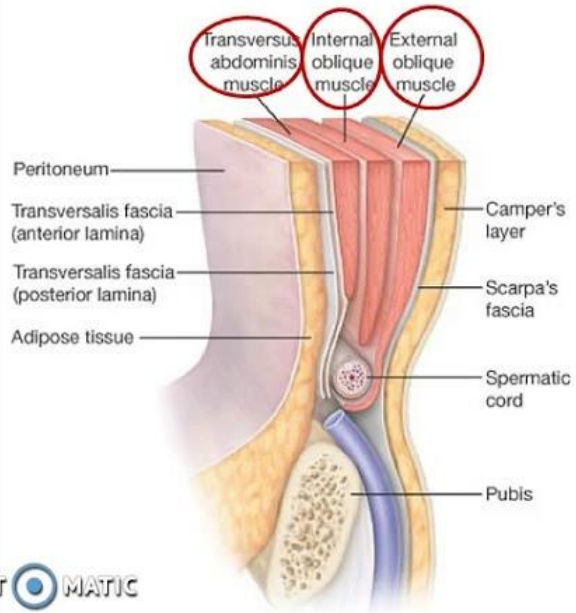
Medially:

- 1- Skin.
- 2- Superficial fascia.
- 3- Rectus abdominis muscle
& its sheath.
- 4- Pyramidalis muscle.
- 5- Fascia transversalis.
- 6- Peritoneum.

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Muscles:

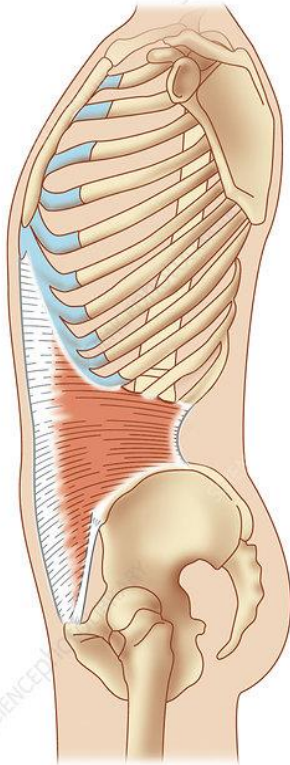
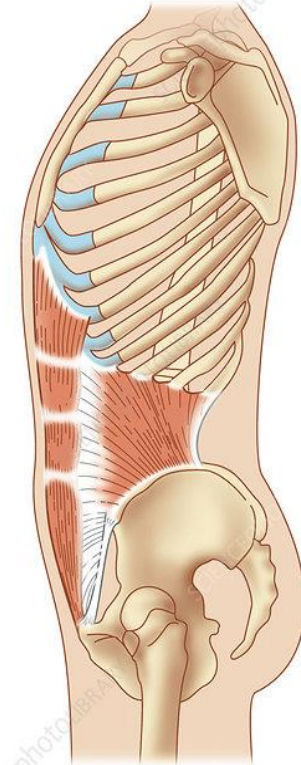
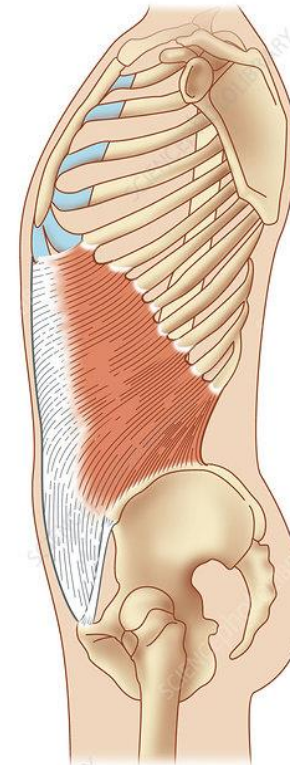
- ✓ **External oblique.**
- ✓ **Internal oblique.**
- ✓ **Transversus abdominis.**



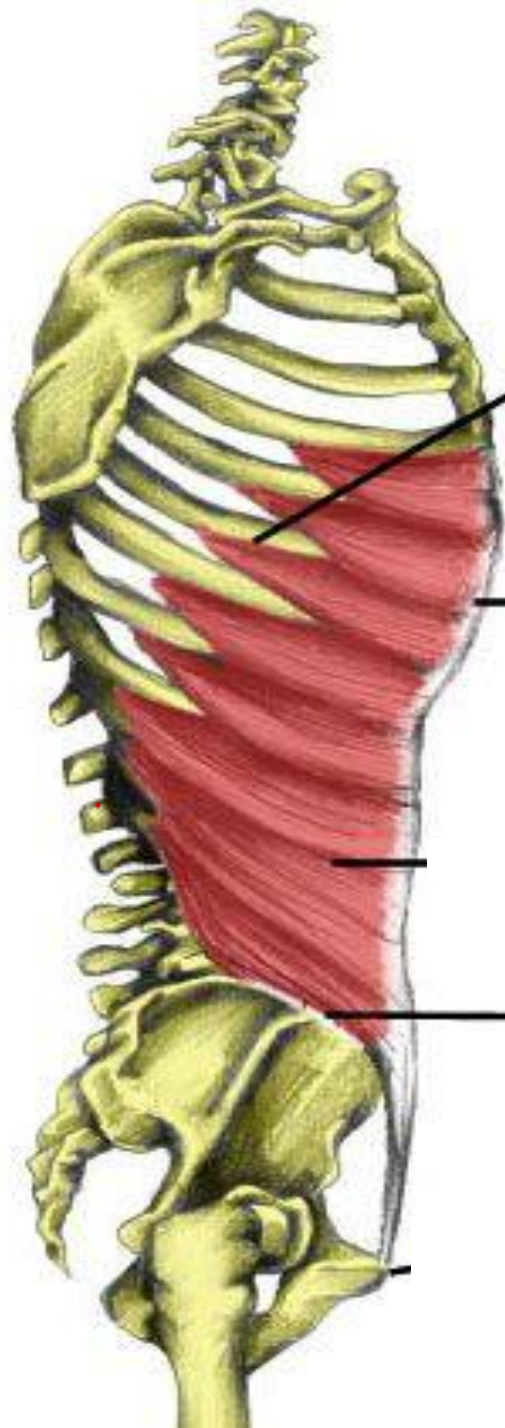
EO

IO

TA



External oblique muscle



Origin Attachment Point

Anterolateral border of the lower 8 ribs

Insertion The muscle is inserted by aponeurosis & fleshy fibers , as follows:

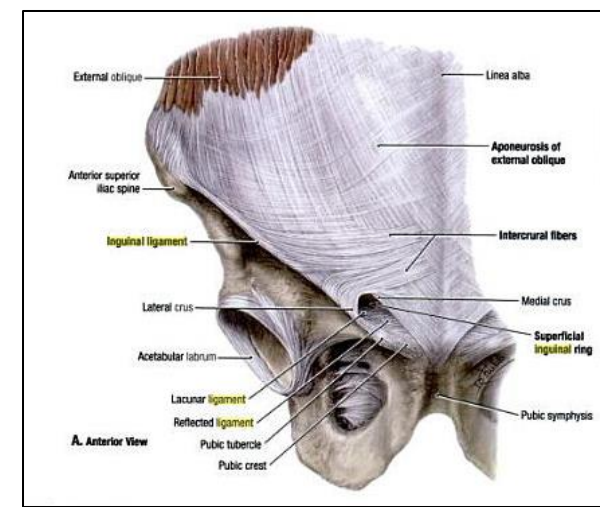
A- Aponeurosis:

1. Medial part → linea alba from xiphoid process to symphysis pubis.

2. Lateral part → lower fibers of aponeurosis is folded **upwards & backwards** upon itself to form the **inguinal ligament** (extends from ASIS → pubic tubercle)

B- Fleshy fibers:

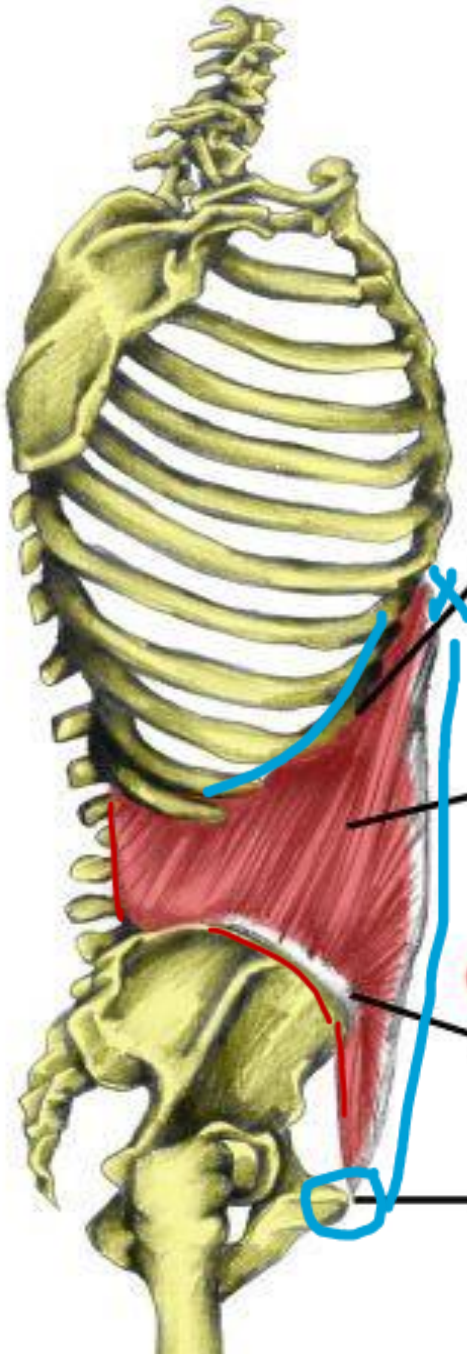
Outer lip of the **iliac crest**.



NS:

Intercostal nerves (T7- T11)
Subcostal nerve (T12)

Internal oblique muscle



Origin

- 1-Lumbar fascia
- 2-Anterior 2/3 of the iliac crest
- 3-The lateral 2/3 of the inguinal ligament

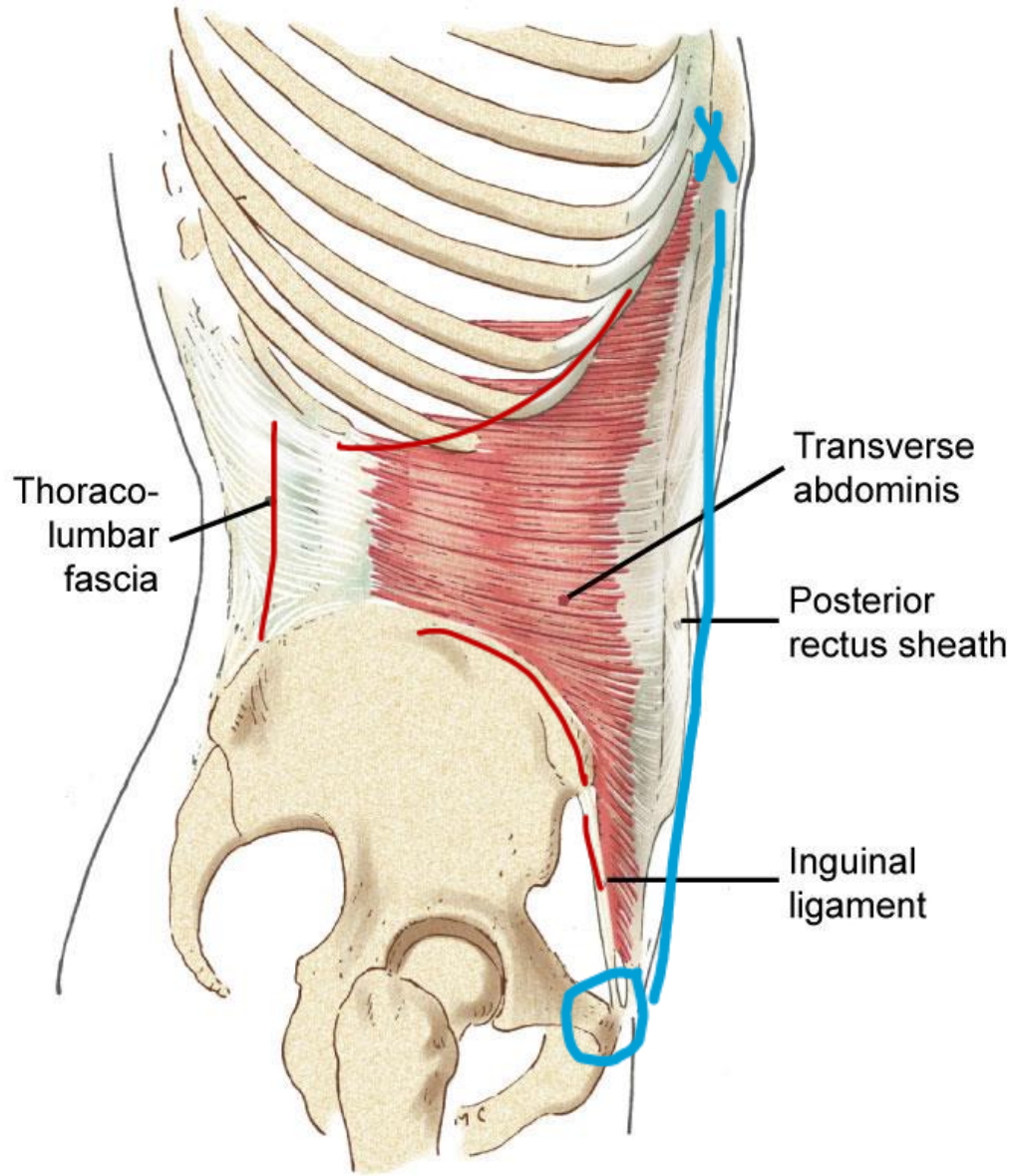
Insertion

- Lower 6 costal cartilages
- Xiphoid process
- Linea Alba
- Pubic crest

NS:

Intercostal nerves (T7- T11)
Subcostal nerve (T12)
Iliohypogastric n.
Ilioinguinal n.

Transverse abdominus muscle



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Origin

- 1-Lower 6 intercostal cartilages.
- 2- Lumbar Fascia
- 3- Ant. 2/3 of iliac crest
- 4- lateral 1/3 of the inguinal ligament

Insertion

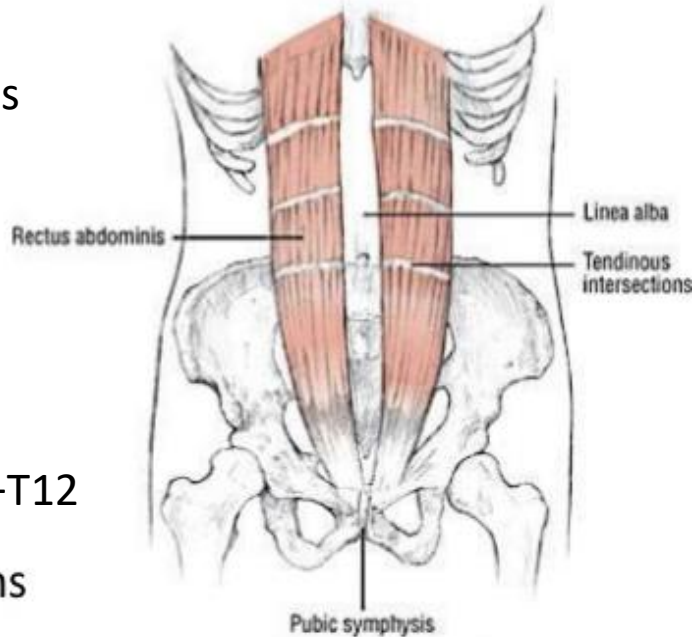
Xiphoid process
Linea Alba
Pubic crest

NS:

Intercostal nerves (T7- T11)
Subcostal nerve (T12)
Iliohypogastric n.
Ilioinguinal n.

Rectus abdominus

- **Origin:** Pubic crest
Pubic symphysis
- **Insertion:** costal
cartilage 5-7, xiphoid
process
- **Direction:** vertical
- **Nerve supply:** T7-T11+T12
- Tendinous intersections
(Divides the muscle into 3 parts)



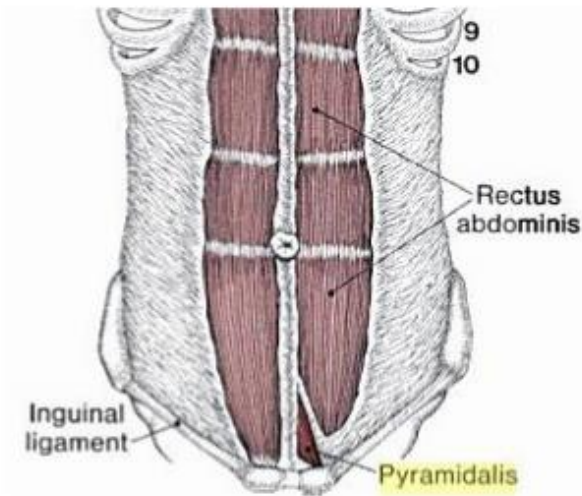
NS:

Intercostal nerves (T7- T11)
Subcostal nerve (T12)

Pyramidalis

- The pyramidalis muscle of the abdomen is a small and triangular muscle, anterior to the rectus abdominis, and contained in the rectus sheath.
- It is absent in 20% of humans and when absent the lower end of the rectus then becomes proportionately increased in size.
- Anatomical studies suggest that the forces generated by the pyramidalis muscles are relatively small

Pyramidalis



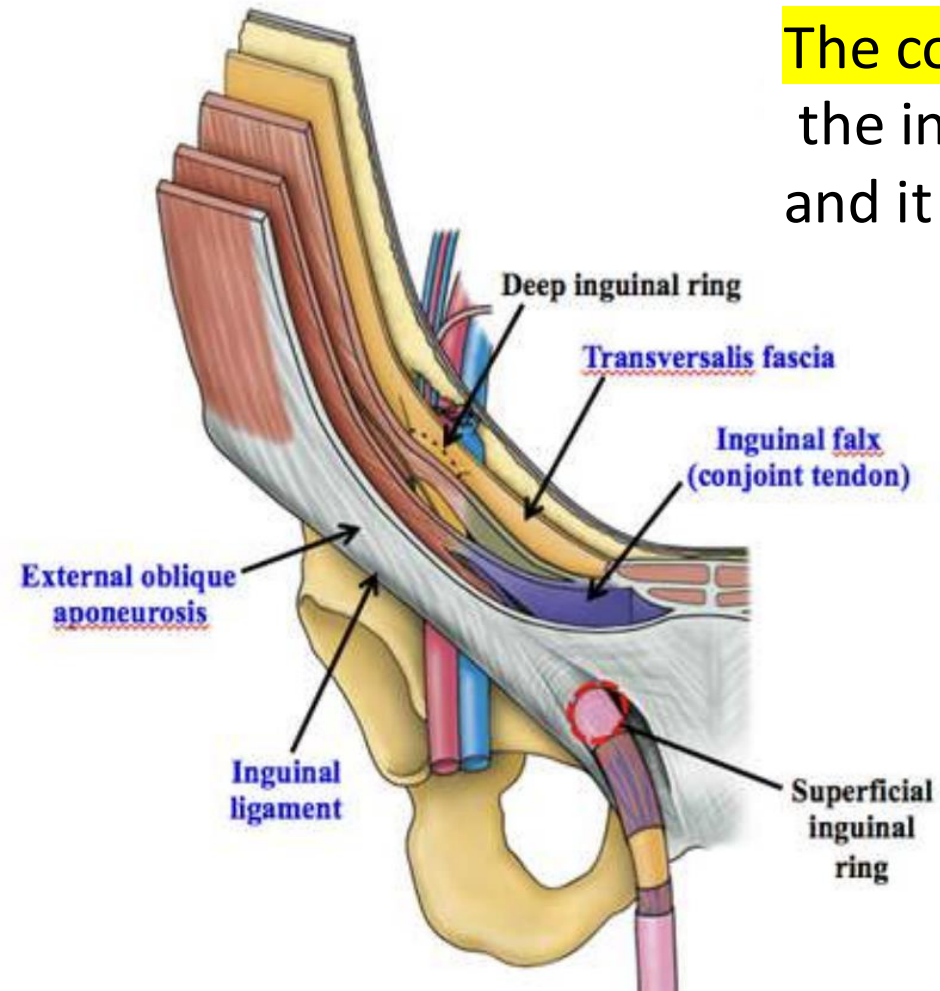
NS:

T12 nerve (subcostal nerve)

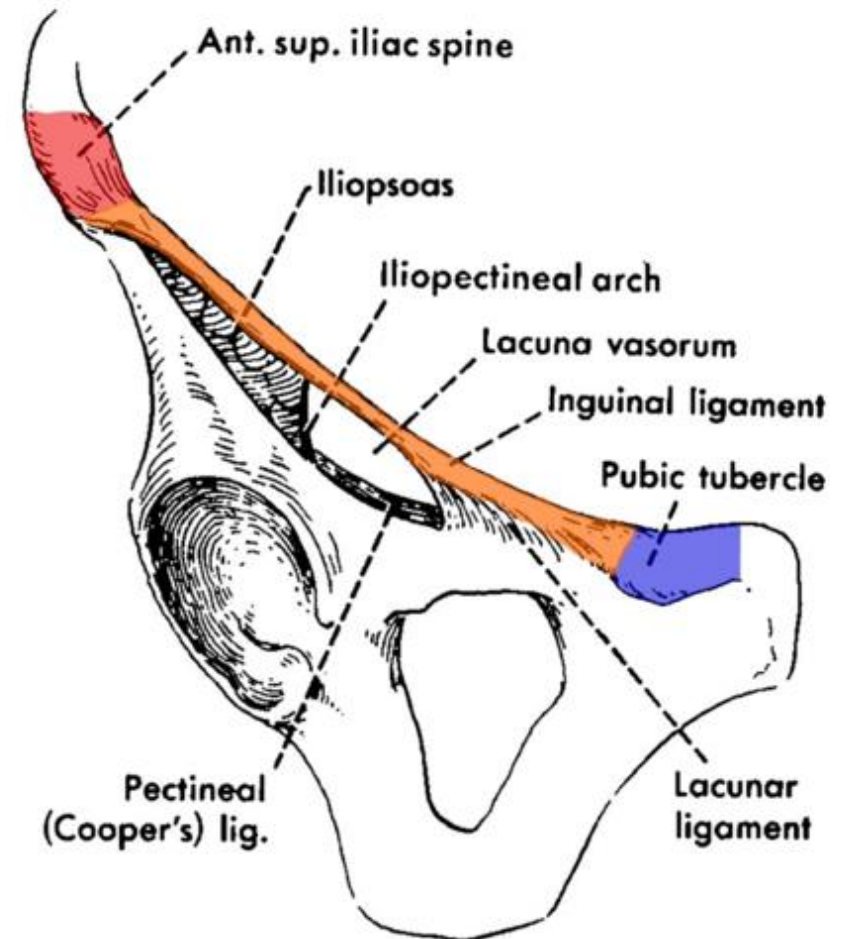
What are the actions of the muscles of the anterior abdominal wall?

- They protect the viscera and help to maintain or to **increase intraabdominal pressure** (help in vomiting – cough – delivery – micturition- defecation).
- They also move the trunk and help to **maintain posture**.
- They are accessory Muscles of **forced Expiration**.
 - **Rectus abdominis**: ventral flexion of the trunk.
 - **2 oblique muscles (external & internal)** bend and rotates the trunk laterally.

The **conjoint tendon** is formed by the fused aponeuroses of: the internal oblique and transversus abdominis muscles, and it inserts into the pubic crest.



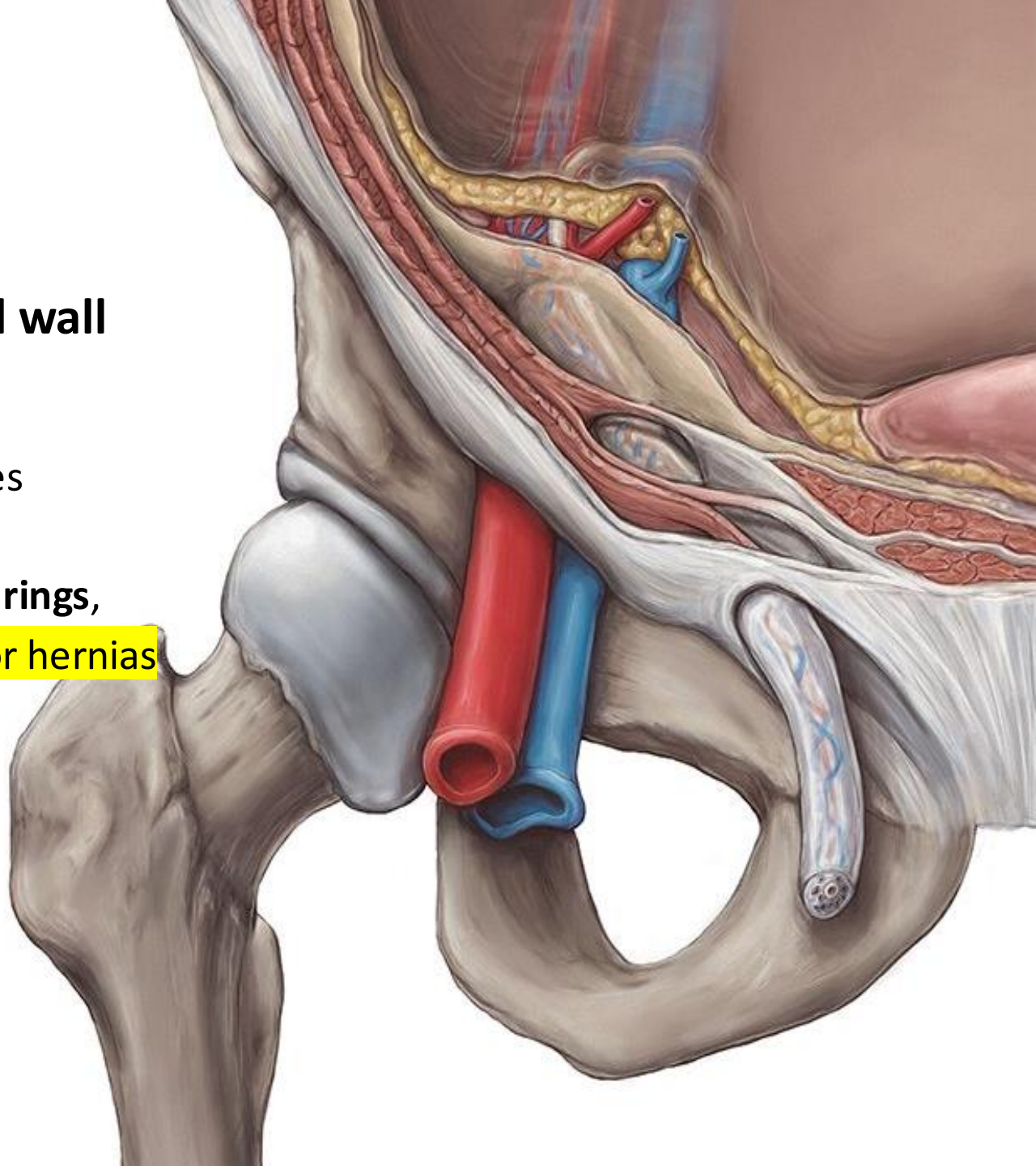
The **lacunar ligament** is a triangular ligament in the groin that attaches the medial end of the inguinal ligament to the pubic tubercle and the pectineal line.



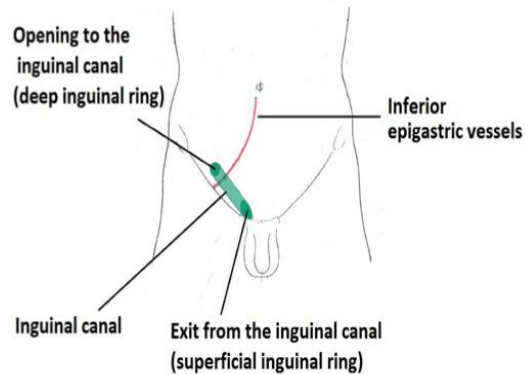
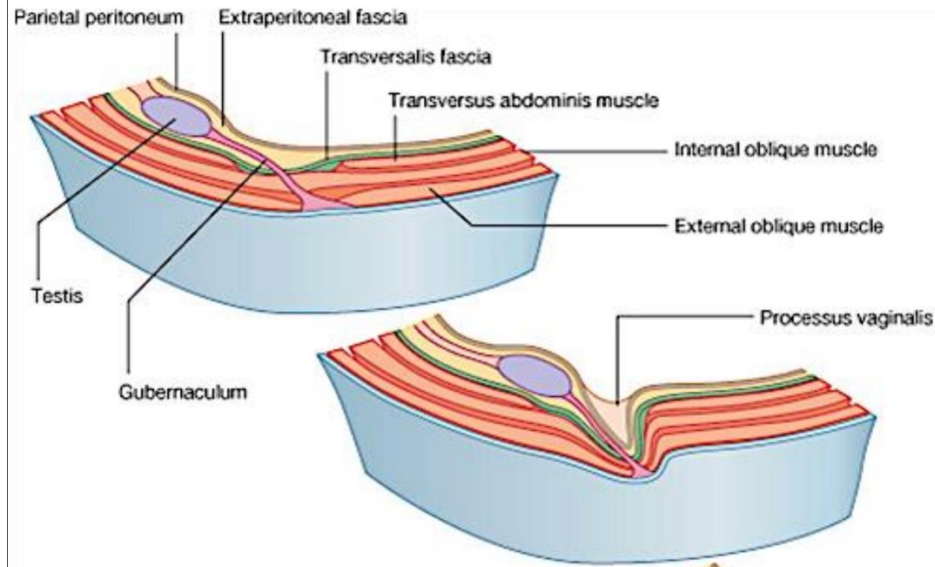
Inguinal canal

- It is located in the **lower anterior abdominal wall** just above the **inguinal ligament**.
- It acts as a short passage for 'spermatic cord in males and 'round ligament of the uterus in females.
- It extends from the deep to the superficial **inguinal rings**,
- This area is clinically significant as a **common site for hernias** due to a natural weakness in the abdominal wall.

- How is it formed?
- Where are the opening ?
- What are its boundaries?
- What are the contents ?



Inguinal canal



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Inferior epigastric artery (position)

Deep Inguinal ring (position)

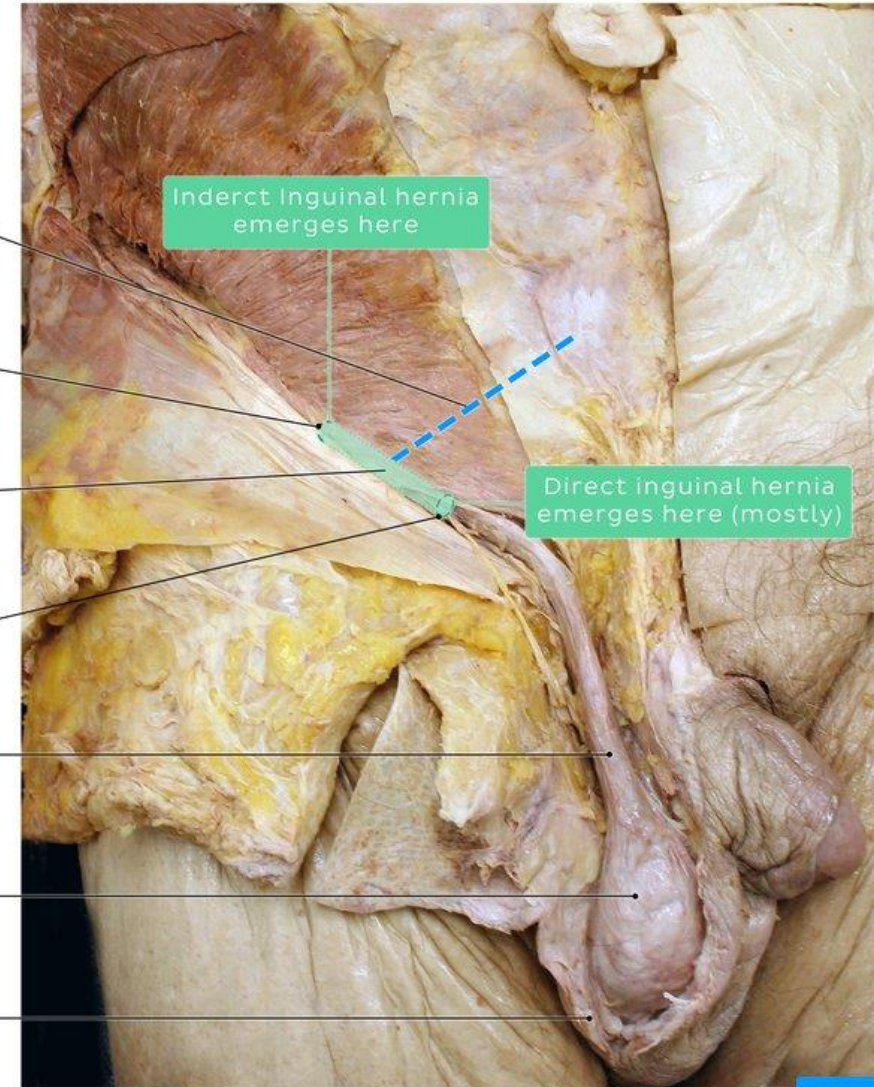
Inguinal canal

Superficial Inguinal ring (position)

Spermatic cord

Testis

Scrotum



Inguinal canal

The inguinal canal has:

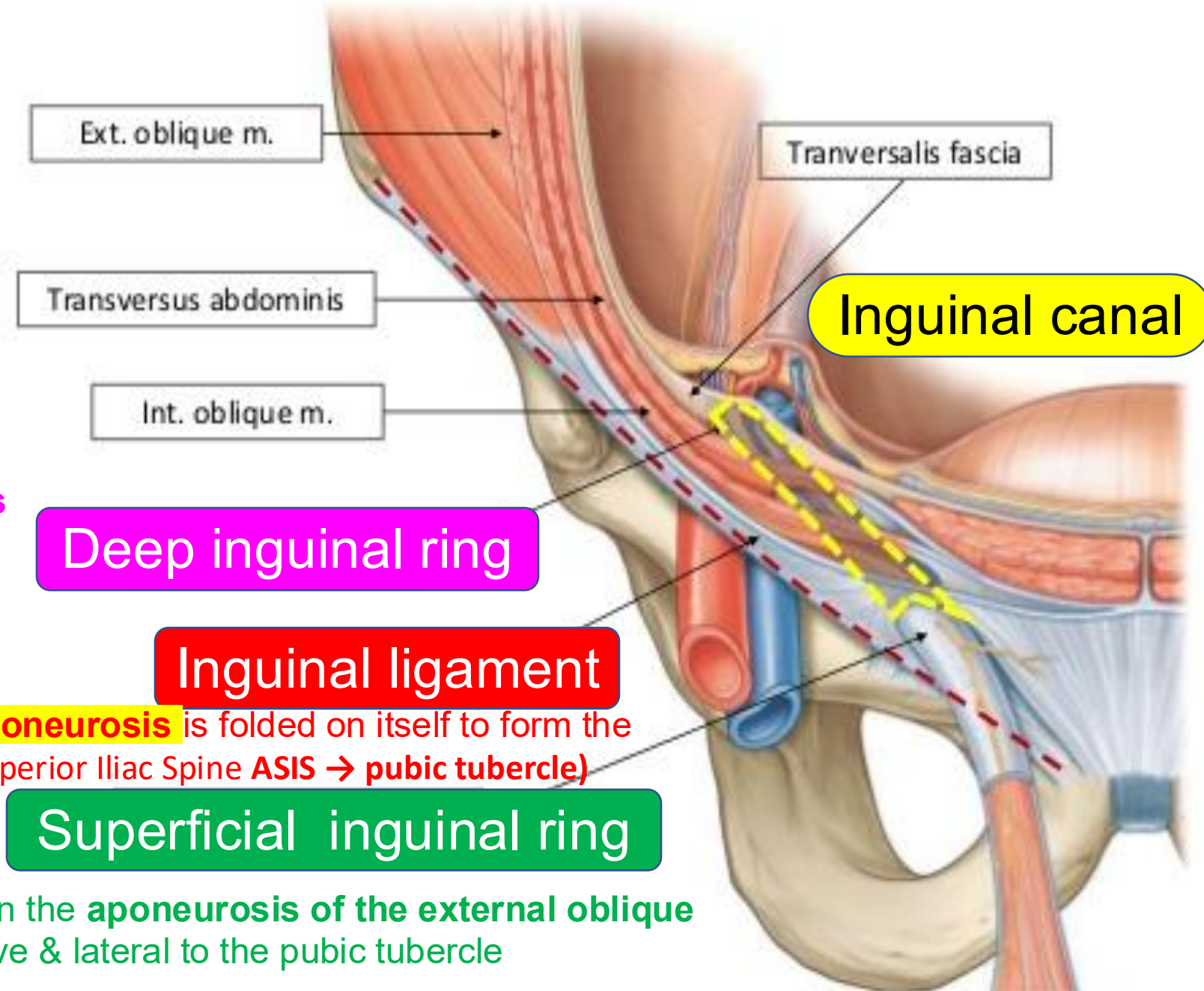
Deep inguinal ring

Superficial inguinal ring

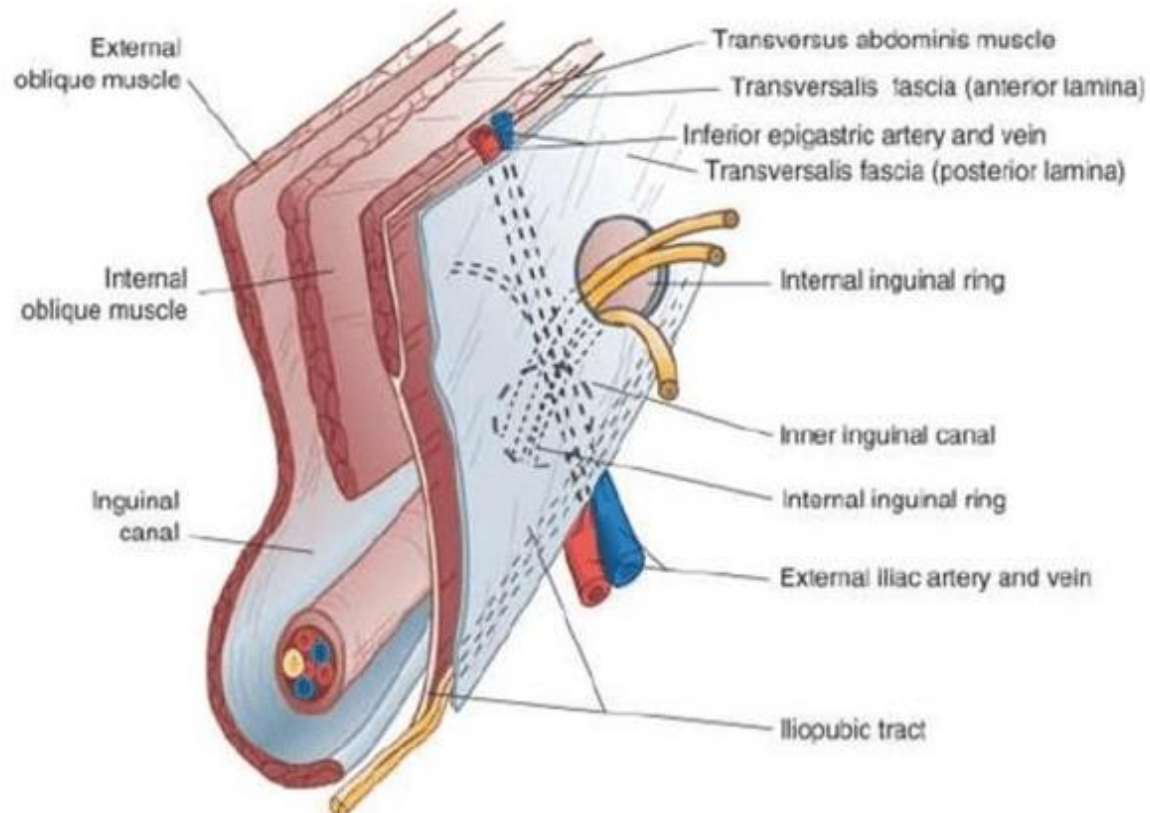
An opening in the **fascia transversalis** (above midpoint of inguinal ligament).

The lower fibers of the **external oblique aponeurosis** is folded on itself to form the **inguinal ligament**. (extends from Anterior Superior Iliac Spine **ASIS** → **pubic tubercle**)

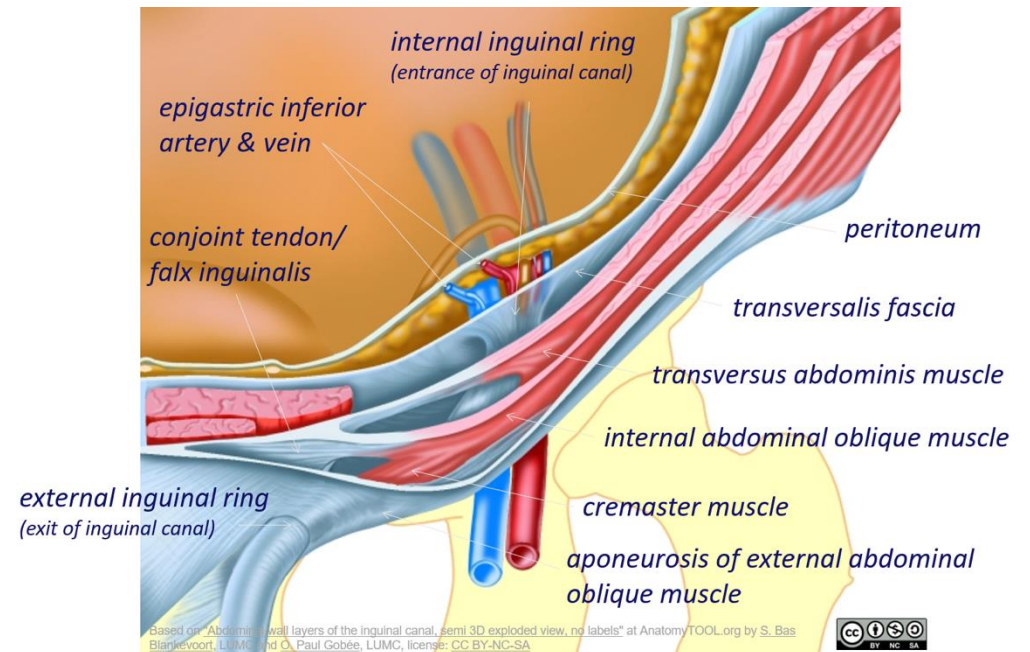
An opening in the **aponeurosis of the external oblique muscle** above & lateral to the pubic tubercle



The Boundaries of the inguinal canal ...



Anterior wall Is derived from Aponeuroses	External oblique aponeurosis, reinforced laterally by origin of internal oblique
Posterior wall	Conjoint tendon medially, fascia transversalis laterally
Roof or superior wall is formed by Muscles	Arching fibers of internal oblique and transversus abdominis
Inferior wall / floor is formed by Ligaments	Inguinal ligament and medially by lacunar ligament



Based on: "Abdominal wall layers of the inguinal canal, semi 3D exploded view, no labels" at AnatomyTOOL.org by S. Bas Blankvoort, LUMC and O. Paul Gobée, LUMC, license: CC BY-NC-SA



The Contents of the inguinal canal ...

Male: spermatic cord and ilioinguinal nerve

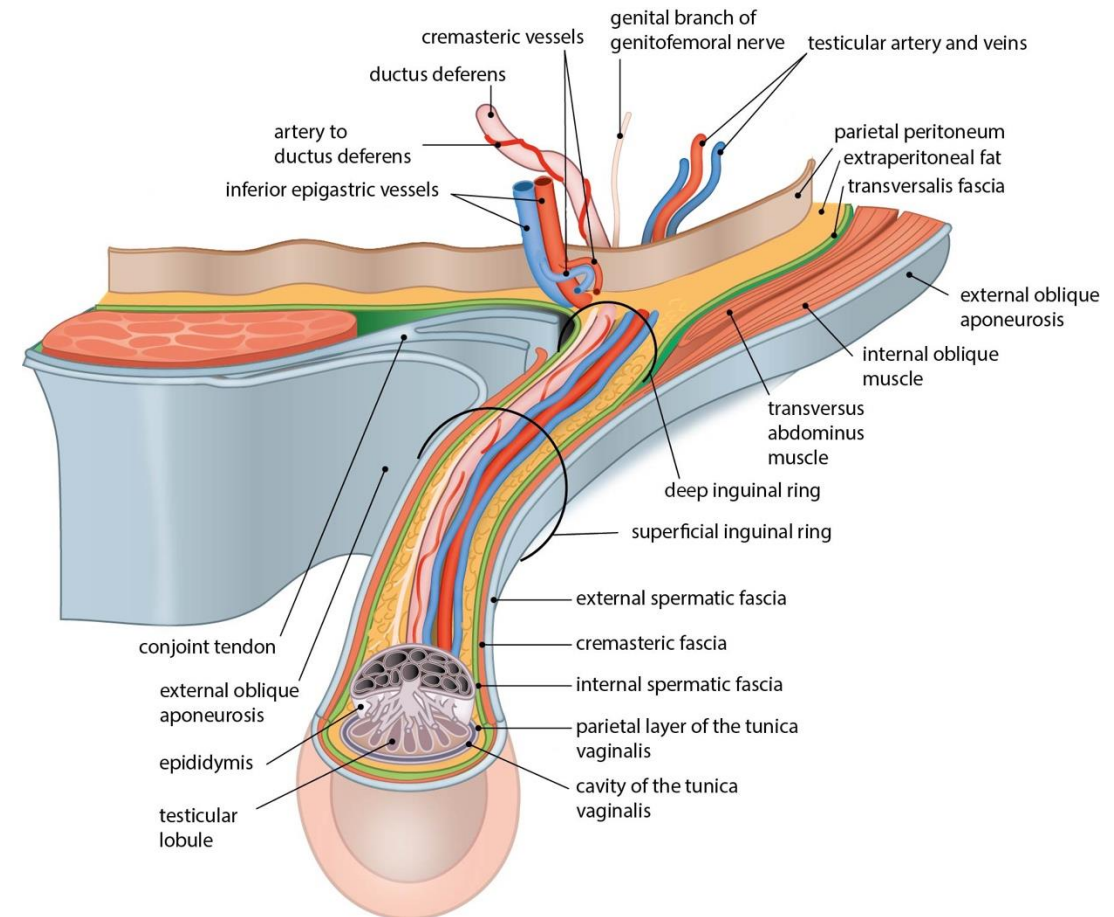
Female: round ligament of the uterus and ilioinguinal nerve

Spermatic cord content (male)

3 fascial layers: external spermatic fascia, cremasteric fascia and muscle, internal spermatic fascia.

3 arteries: vas (ductus) deferens artery, cremasteric artery, testicular artery.

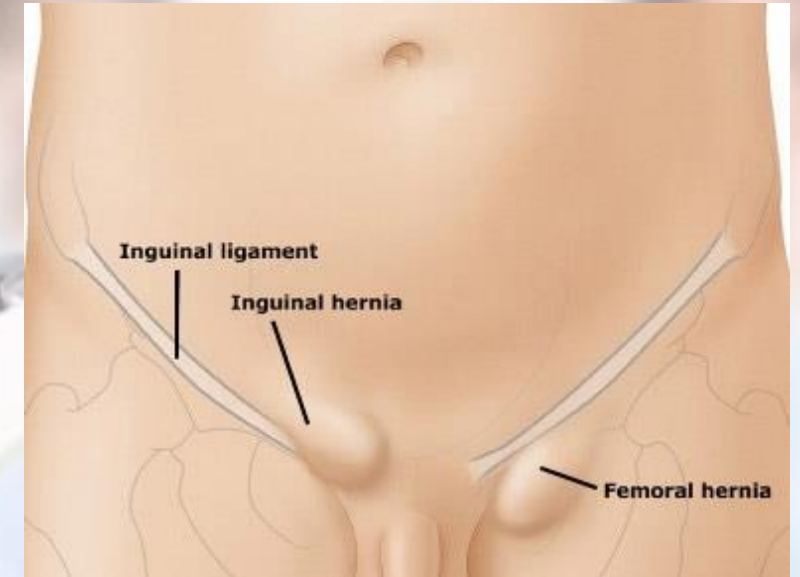
3 nerves: ilioinguinal nerve, genital branch of genitofemoral nerve, sympathetic fibers,



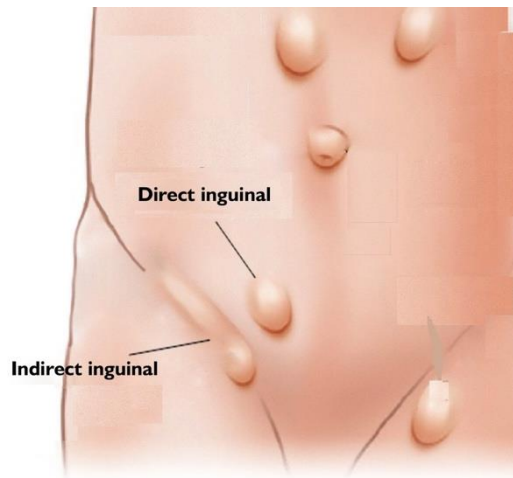
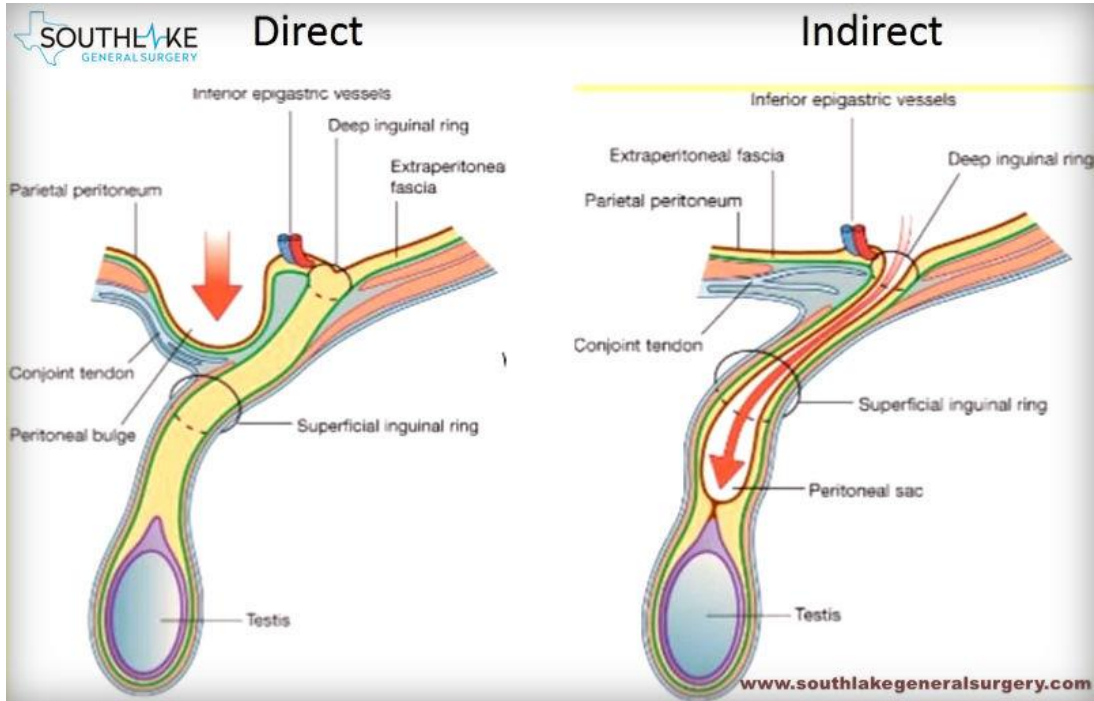
Clinical implications

Hernias:

A hernia is the abnormal protrusion of a portion of the intestine or viscera out of the walls of its containing cavity as a result of a weakness in these walls.



What are the Types of Inguinal Hernias



Indirect inguinal hernias: more common

indirect hernias are **congenital**,

Cause: due to failure of the deep inguinal ring to close before birth.

It follows the spermatic cord path through the deep & superficial inguinal rings.

Can extend into the scrotum

Age: More common in younger males & infants.

Indirect inguinal hernias:

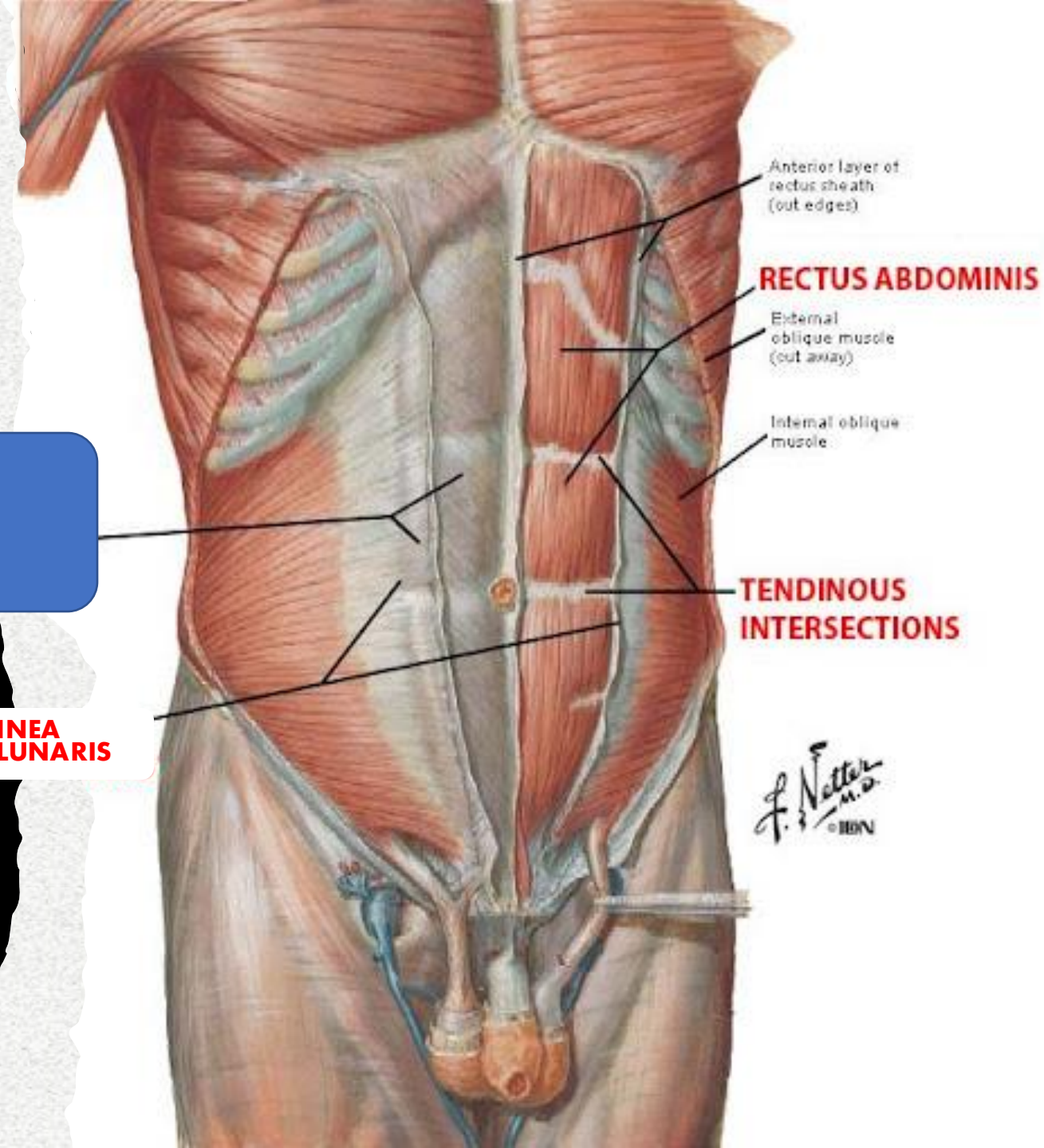
Direct hernias are **acquired**,

Cause: due to weakness in 'abdominal wall. It protrudes through the posterior wall of the inguinal canal without going through 'deep ring.

Age: More common in middle-aged and older men.

Rectus sheath

**LINEA
SEMILUNARIS**



- S = Skin
- SF = Subcutaneous fat
- RA = Rectus abdominis
- EO = External oblique
- IO = Internal oblique
- TO = Transversus abdominis
- TF = Transversalis fascia
- EF = Extraperitoneal fascia
- P = Peritoneum

● Anastomoses of lower intercostal, subcostal, lumbar and ascending branch of deep circumflex vessels

What is the Linea Alba
What is the Linea semilunaris

What are the exact levels ?

Above the costal margin

From ' costal margin
To midway between umbilicus & symphysis pubic.

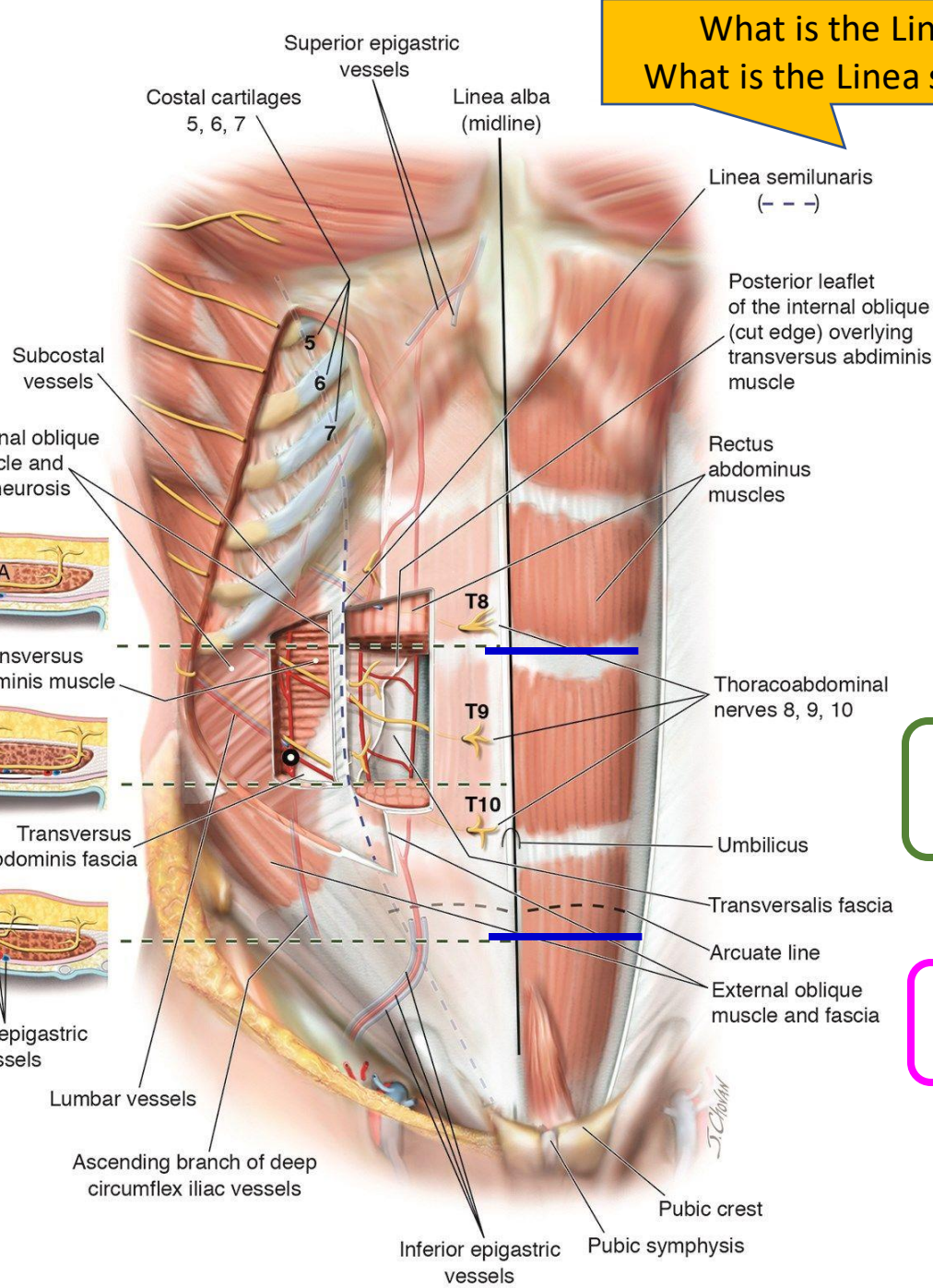
From midway between umbilicus & symphysis pubic
To symphysis pubic

Upper 1/3

Middle 1/3

Lower 1/3

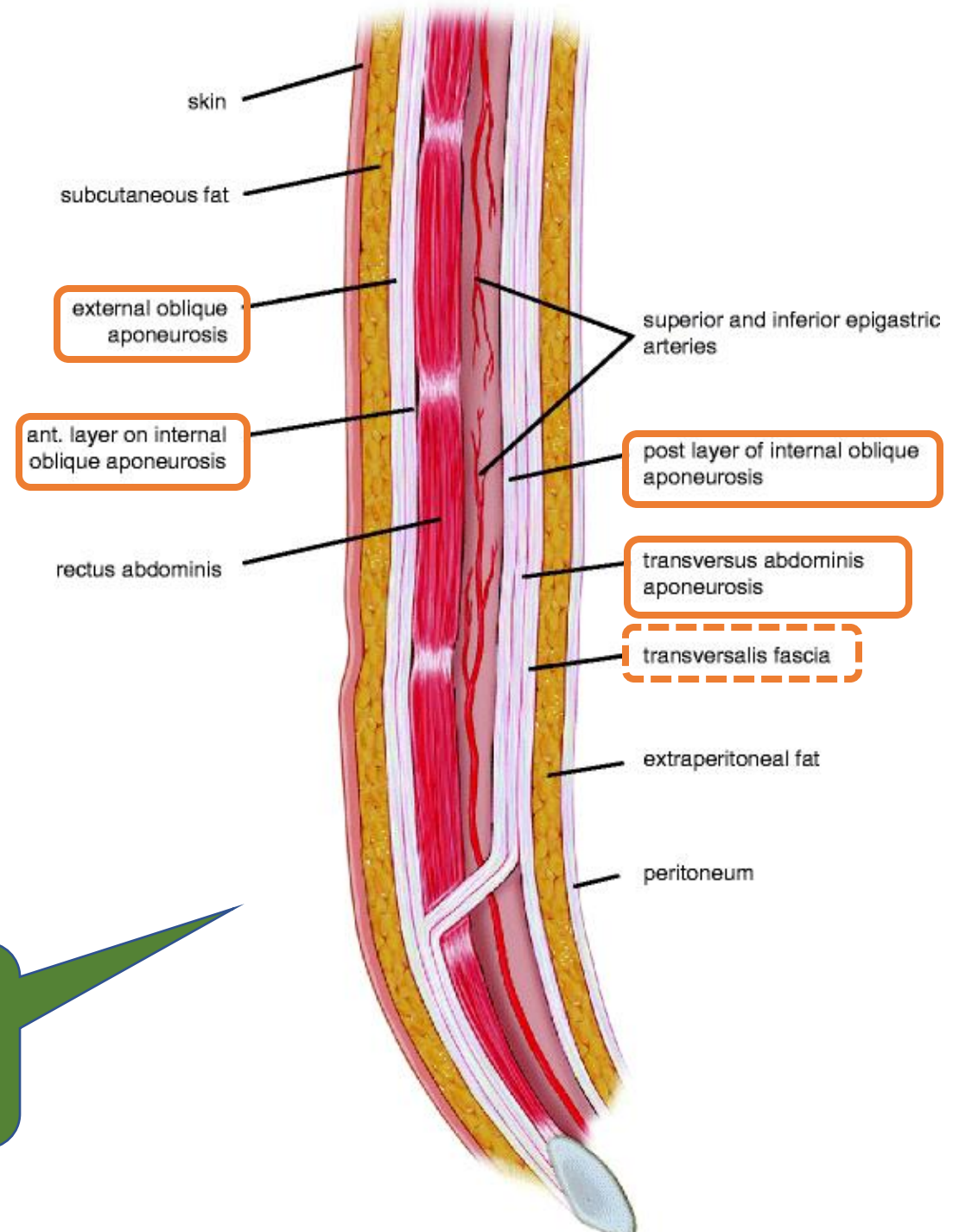
Neurovascular plane:
Between internal oblique and transversus abdominis muscles



Rectus sheath

How is it formed ?

What are the contents of the rectus sheath?



Rectus sheath

The rectus sheath is a tendon sheath (aponeurosis) which encloses the [rectus abdominis](#) and [pyramidalis muscles](#). It is an extension of the aponeurosis of the [external abdominal oblique](#), [internal abdominal oblique](#), and [transversus abdominis](#) muscles.

How is it formed?

It has anterior and posterior **walls**, according to **different levels**.
See pic.

What are the contents?

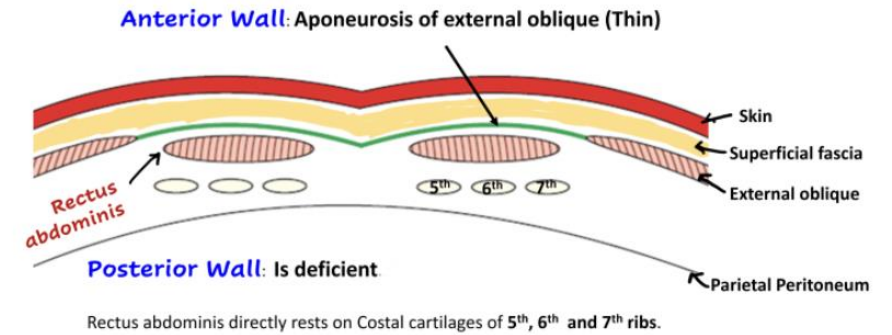
Muscles: [rectus abdominis](#) and [pyramidalis muscles](#)

Nerves: lower 5 intercostal nerves and subcostal nerve (T7-T12)

Blood vessels: superior and inferior epigastric blood vessels.

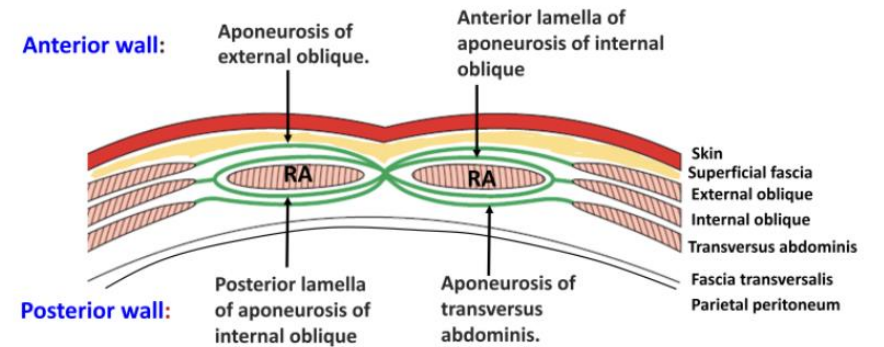
In the region above the costal margin

Upper 1/3



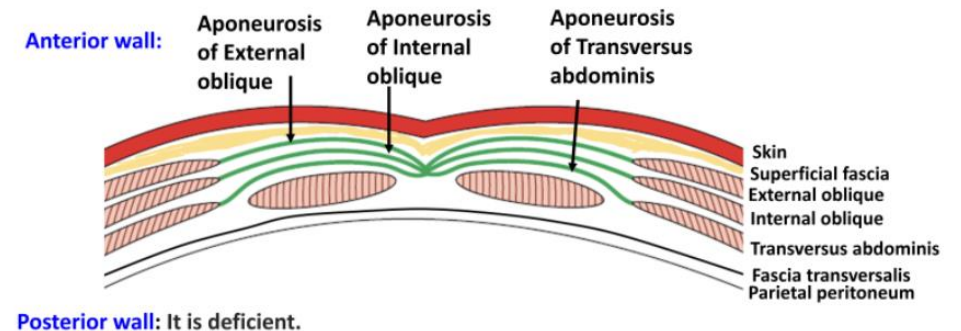
In the region from costal margin to midway between umbilicus and pubic symphysis

Middle 1/3



In the region between the midpoint of umbilicus and pubic symphysis to pubic symphysis

Lower 1/3



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Thank you